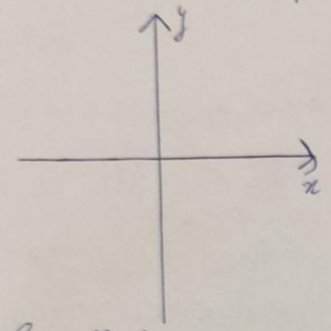
Бесконечные односторон...

1) $\lim\limits_{x\to a+0} f(x)=\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D$

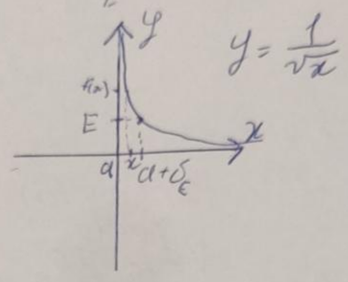
$a<x<a+\delta_E \Rightarrow |f(x)|>E$



2) $\lim\limits_{x\to a+0} f(x)=+\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D$

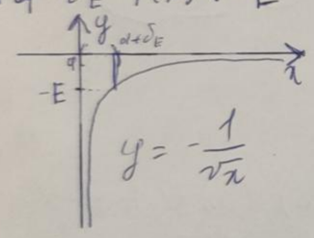
$a<x<a+\delta_E \Rightarrow f(x)>E$



3) $\lim\limits_{x\to a+0} f(x)=-\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D$

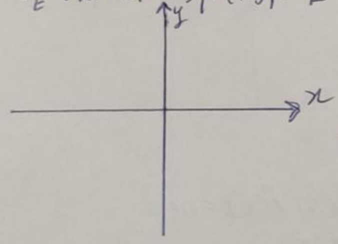
$a<x<a+\delta_E\quad f(x)<-E$



4) $\lim\limits_{x\to a-0} f(x)=\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D:$

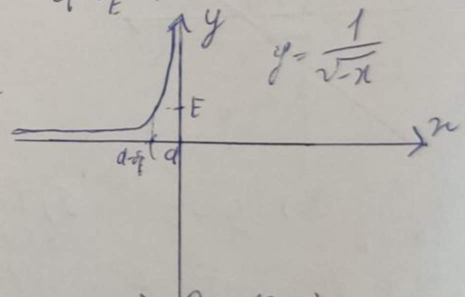
$a-\delta_E<x<a \Rightarrow |f(x)|>E$



5) $\lim\limits_{x\to a-0} f(x)=+\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D.$

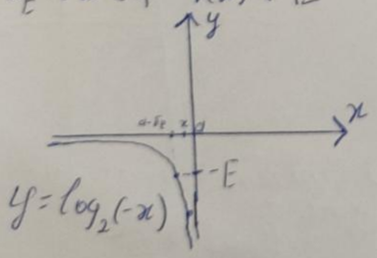
$a-\delta_E<x<a\quad f(x)>E$



6) $\lim\limits_{x\to a-0} f(x)=-\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D$

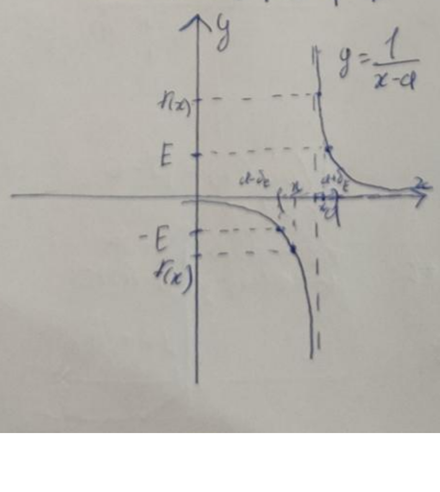
$a-\delta_E<x<a\quad f(x)<-E$



7) $\lim\limits_{x\to a} f(x)=\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D$

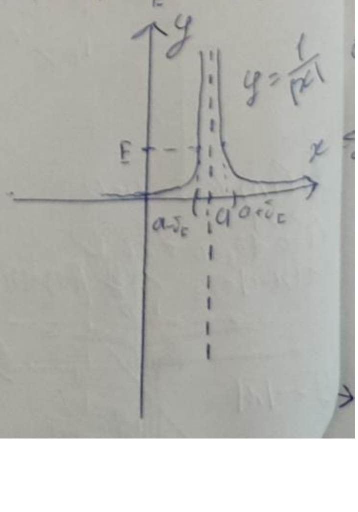
$0<|x-a|<\delta_E\quad |f(x)|>E$



8) $\lim\limits_{x\to a} f(x)=+\infty$

$\forall E>0\ \exists \delta_E>0\ \forall x\in D$

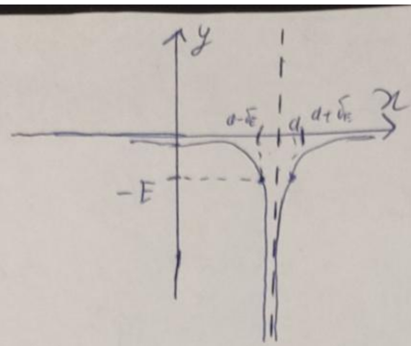
$0<|x-a|<\delta_E\quad f(x)>E$



9) $\lim\limits_{x \to a} f(x) = -\infty$

$\forall E > 0 \ \exists \delta_E > 0 \ x \in D$

$0 < |x-a| < \delta_E \quad f(x) < -E$



$y = \dfrac{-1}{|x-a|}$

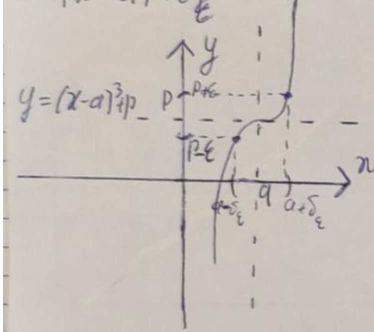
Конечные односторонние пределы

1) $\lim\limits_{x \to a} f(x) = p$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \ \forall x \in D$

$0 < |x-a| < \delta_E \quad |f(x) - p| < \varepsilon$

$y = (x-a)^3 + p$

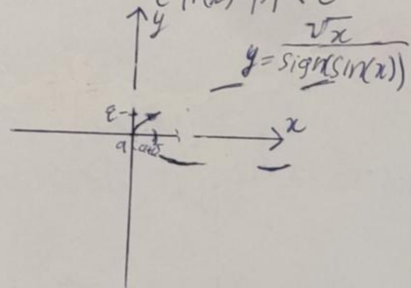


2) $\lim\limits_{x \to a+0} f(x) = p$

$\forall \varepsilon > 0 \ \exists \delta_E > 0 \ \forall x \in D$

$a < x < a + \delta_\varepsilon \ |f(x) - p| < \varepsilon$

$y = \dfrac{\sqrt{x}}{\mathrm{sign}(\sin(x))}$

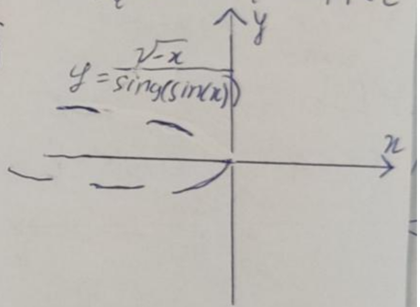


3) $\lim\limits_{x \to a-0} f(x) = p$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \ \forall x \in D$

$a - \delta_\varepsilon < x < a \ |f(x) - p| < \varepsilon$

$y = \dfrac{\sqrt{-x}}{\mathrm{sing}(\sin(x))}$

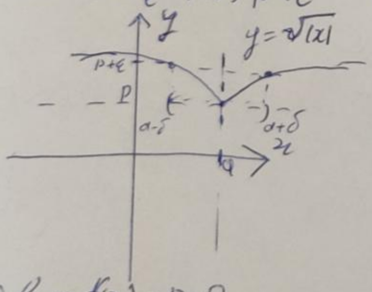


4) $\lim\limits_{x \to a} f(x) = p + 0$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \ \forall x \in D$

$0 < |x-a| < \delta_\varepsilon \quad f(x) - p < \varepsilon$

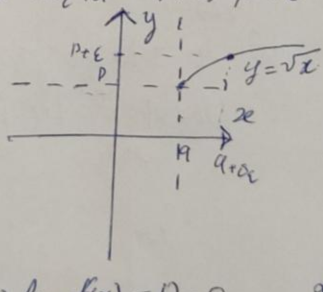
$y = \sqrt{|x|}$



5) $\lim\limits_{x \to a+0} f(x) = p + 0$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \ \forall x \in D$

$a < x < \delta_\varepsilon + a \quad f(x) - p < \varepsilon$

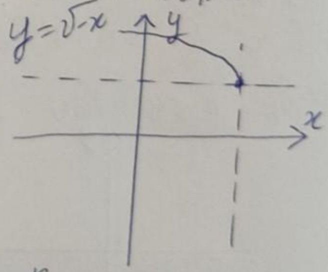


6) $\lim\limits_{x \to a-0} f(x) = p + 0$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \ \forall x \in D$

$a - \delta_\varepsilon < x < a \quad f(x) - p < \varepsilon$

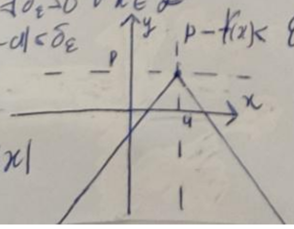
$y = \sqrt{-x}$



7) $\lim\limits_{x \to a} f(x) = p - 0$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \ \forall x \in D$

$0 < |x-a| < \delta_\varepsilon \quad p - f(x) < \varepsilon$

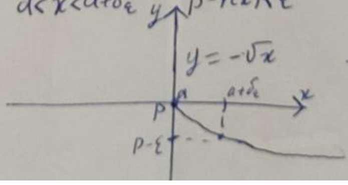


$y = -|x|$

8) $\lim\limits_{x \to a+0} f(x) = p - 0$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \ x \in D$

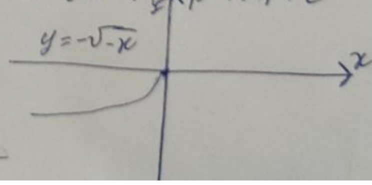
$a < x < a + \delta_\varepsilon \quad p - f(x) < \varepsilon$



9) $\lim\limits_{x \to a-0} f(x) = p - 0$

$\forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0 \quad x \in D$

$a - \delta_\varepsilon < x < a \quad p - f(x) < \varepsilon$

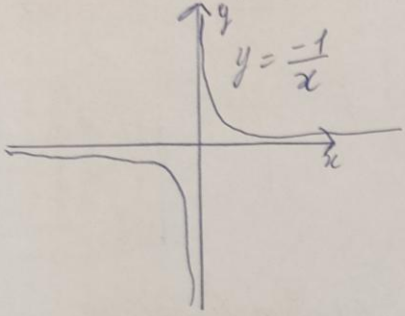


Конечные пределы в бесконечности

1) $\lim\limits_{x \to \infty} f(x) = p$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

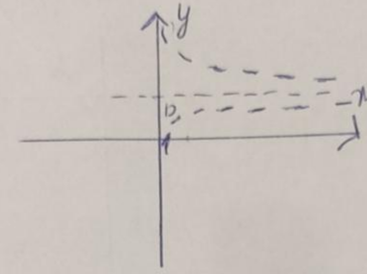
$|x| > \Delta_\varepsilon \quad |f(x) - p| < \varepsilon$



2) $\lim\limits_{x \to +\infty} f(x) = p$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

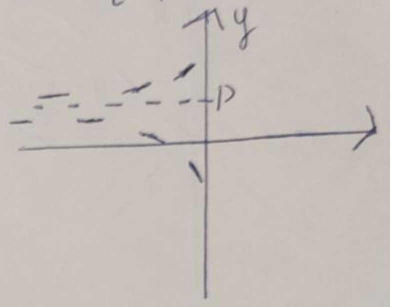
$x > \Delta_\varepsilon \quad |f(x) - p| < \varepsilon$



3) $\lim\limits_{x \to -\infty} f(x) = p$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

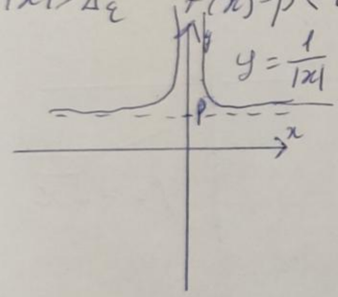
$x < -\Delta_\varepsilon \quad |f(x) - p| < \varepsilon$



4) $\lim\limits_{x \to \infty} f(x) = p + 0$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

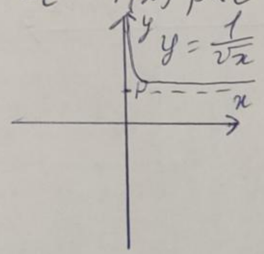
$|x| > \Delta_\varepsilon \quad f(x) - p < \varepsilon$



5) $\lim\limits_{x \to +\infty} f(x) = p + 0$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

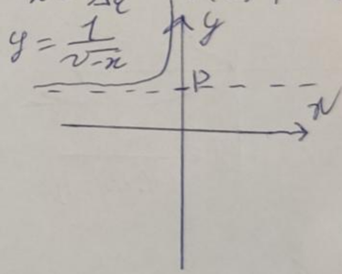
$x > \Delta_\varepsilon \quad f(x) - p < \varepsilon$



6) $\lim\limits_{x \to -\infty} f(x) = p + 0$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

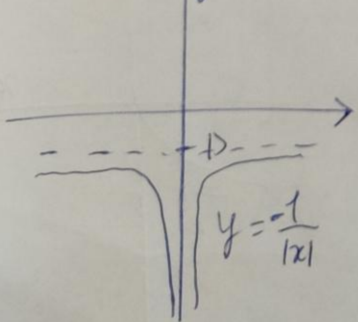
$x < -\Delta_\varepsilon \quad f(x) - p < \varepsilon$



7) $\lim\limits_{x \to \infty} f(x) = p - 0$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

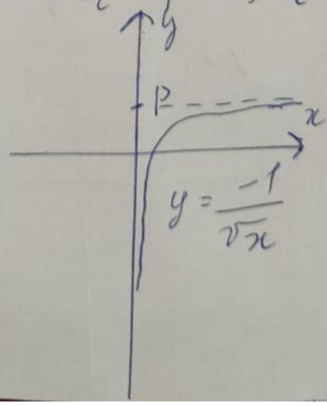
$|x| > \Delta_\varepsilon \quad p - f(x) < \varepsilon$



8) $\lim\limits_{x \to +\infty} f(x) = p - 0$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

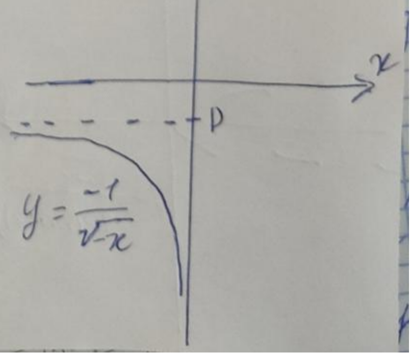
$x > \Delta_\varepsilon \quad p - f(x) < \varepsilon$



9) $\lim\limits_{x \to -\infty} f(x) = p - 0$

$\forall \varepsilon > 0 \ \exists \Delta_\varepsilon > 0 \ \forall x \in D$

$x < -\Delta_\varepsilon \quad p - f(x) < \varepsilon$

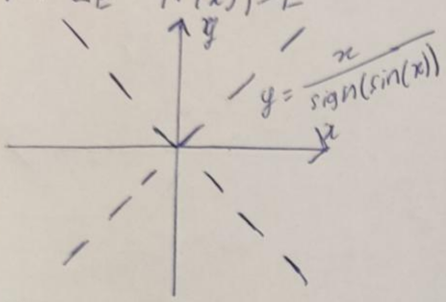


Бесконечные пределы в бесконечных точках

1) $\lim\limits_{x\to\infty} f(x)=\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

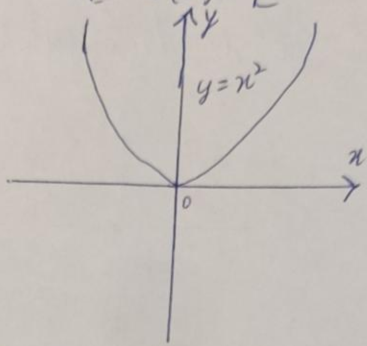
$|x|>\Delta_E \quad |f(x)|>E$



2) $\lim\limits_{x\to\infty} f(x)=+\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

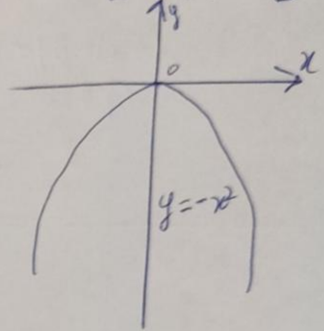
$|x|>\Delta_E \quad f(x)>E$



3) $\lim\limits_{x\to\infty} f(x)=-\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

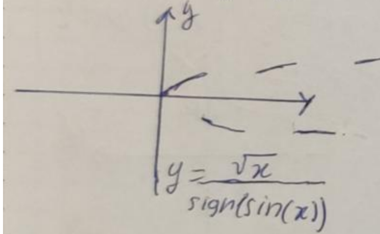
$|x|>\Delta_E \quad f(x)<-E$



4) $\lim\limits_{x\to+\infty} f(x)=\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

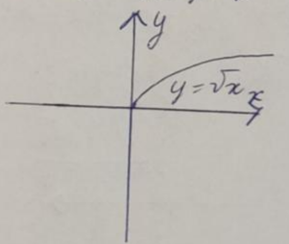
$x>\Delta_E \quad |f(x)|>E$



5) $\lim\limits_{x\to+\infty} f(x)=+\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

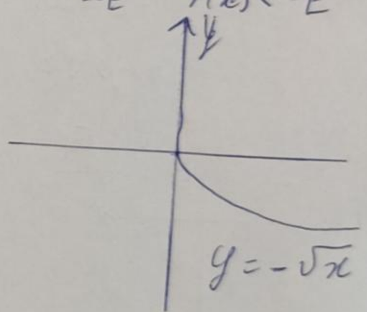
$x>\Delta_E \quad f(x)>E$



6) $\lim\limits_{x\to+\infty} f(x)=-\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

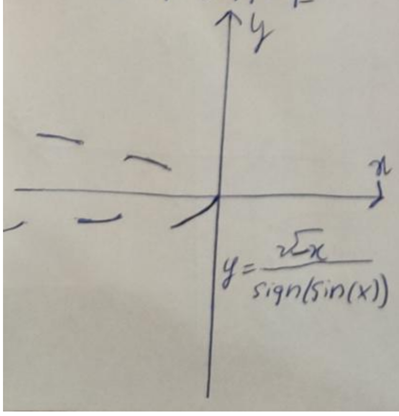
$x>\Delta_E \quad f(x)<-E$



7) $\lim\limits_{x\to-\infty} f(x)=\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

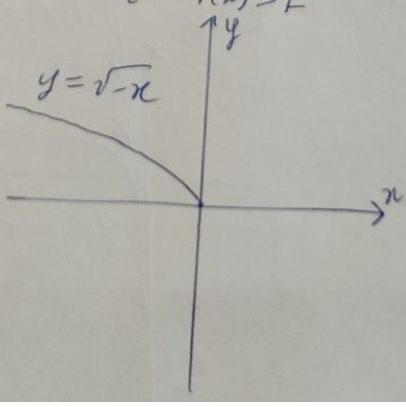
$x<-\Delta_E \quad |f(x)|>E$



8) $\lim\limits_{x\to-\infty} f(x)=+\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

$x<-\Delta_E \quad f(x)>E$



9) $\lim\limits_{x\to-\infty} f(x)=-\infty$

$\forall E>0\ \exists \Delta_E>0\ \forall x\in \mathcal{D}$

$x<-\Delta_E \quad f(x)<-E$